

A Comparative Study of Classical Machine Learning and Convolutional Neural Networks under Different Training Data Sizes and Noise Levels

Dang Khoa Duong
khoa.dd2516142@sis.hust.edu.vn
Hanoi University of Science and Technology (HUST)
Hanoi, Vietnam

Abstract

This study compares a classical machine learning model and a convolutional neural network under different training data sizes and levels of image noise. We first examine how the performance of the two models changes as the amount of training data increases. We then add increasing levels of noise to the test images to study how strongly each model is affected. The results show that the classical machine learning model performs better when only a small amount of training data is available, while the convolutional neural network performs better when the full training set is used. Under mild noise, the classical machine learning model shows a much larger accuracy drop than the convolutional neural network. Although the convolutional neural network shows a larger overall decrease across the full noise range, this is mainly related to its higher starting performance. These results show that model performance depends on both data availability and noise level.

1 Introduction

Machine learning models can perform differently depending on the amount and quality of available data. Classical machine learning methods can work well with limited training data, while convolutional neural networks can learn useful features directly from images when enough data are available. However, it is not clear how the relative performance of these two types of models changes as the amount of training data increases.

Their performance may also change when test images contain different levels of noise. Understanding these changes is important for choosing a suitable model under different data conditions.

This study compares a classical machine learning model and a convolutional neural network under different training data sizes and noise levels. First, we examine how their performance changes when different amounts of training data are used. Then, we evaluate how their accuracy changes as image noise increases. The study uses statistical analysis to determine whether the observed differences are meaningful and to verify two hypotheses about data size and noise robustness.

2 Background

2.1 Classical Machine Learning

Classical machine learning models learn a relationship between input features and target labels from training data. In image classification, the input is usually represented by a set of numerical features extracted from each image. These models learn from the provided features rather than automatically learning complex image representations. Their performance can therefore depend strongly on the quality of feature representation and the amount of training data available.

A major advantage of classical machine learning is that many models can achieve reasonable performance with relatively small training sets. However, their ability to capture complex patterns may be limited when the in-

put data contain rich spatial information. This difference is important when comparing classical machine learning with models that can learn image features directly.

2.2 Convolutional Neural Networks

A convolutional neural network is a deep learning model designed to process data with a spatial structure, such as images. Instead of relying on manually designed features, it learns useful image features directly from the training data. Convolutional layers apply small filters to local regions of an image and gradually learn patterns such as edges, shapes, and more complex structures. Later layers combine these patterns to support the final classification.

Because the model learns its own features, a convolutional neural network can benefit from a larger amount of training data. However, when only a small amount of data is available, the model may not have enough information to learn useful features effectively. This provides a basis for studying how its performance changes as the training data size increases.

2.3 Image Noise and Model Robustness

Image noise refers to unwanted changes in pixel values that make an image different from its original form. In this study, noise is added to test images at different levels to examine how the two models respond to corrupted inputs. As the noise level increases, the input becomes less similar to the data seen during training, which may reduce prediction accuracy.

Model robustness describes how well a model can maintain its performance when the input contains such disturbances. Different models may respond differently to noise because they use different ways of representing and processing the input. Therefore, measuring the change in accuracy across several noise levels allows us to compare the robustness of the classical machine learning model and the convolutional neural network.

3 Methodology

3.1 Research Questions

This study addresses two research questions:

RQ₁: *How does the performance of a classical machine learning model and a convolutional neural network change*

as the amount of training data increases?

RQ₂: *How does increasing image noise affect the performance of a classical machine learning model and a convolutional neural network?*

3.2 Hypotheses

3.2.1 Hypothesis 1: Data Size Scaling

When only a small portion of the training data is available, the classical machine learning model and the convolutional neural network are expected to achieve similar performance. When more than 30% of the training data is used, the convolutional neural network is expected to outperform the classical machine learning model.

The hypothesis is considered unsupported if the difference between the two models exceeds the predefined range in the low-data setting, or if the convolutional neural network gains a performance advantage greater than the predefined threshold in the high-data setting.

3.2.2 Hypothesis 2: Noise Robustness

As the level of image noise increases, the convolutional neural network is expected to experience a faster decrease in accuracy than the classical machine learning model. The hypothesis is based on the assumption that noise affecting the input can propagate through the multiple layers of the convolutional neural network and lead to a larger loss in performance.

3.3 Classification Task

The task is an image classification problem in which each input image is assigned to one of a predefined set of classes. The same dataset and evaluation procedure are used for both models so that their performance can be compared under the same conditions.

The study focuses on two factors that may affect classification performance: the amount of training data and the level of noise in the test images. The first experiment changes the amount of training data, while the second experiment keeps the trained models fixed and changes the level of noise in the test data.

3.4 Compared Approaches

This study compares two approaches for image classification: a classical machine learning approach and a convo-

lutional neural network approach. Both approaches use the same training and test data, but they process the images differently.

The classical machine learning approach consists of three classifiers: Linear Support Vector Machine, Logistic Regression, and Random Forest. Before classification, image features are extracted using the Histogram of Oriented Gradients method. The extracted features are then standardized and reduced to 50 principal components using Principal Component Analysis before being given to each classifier.

The convolutional neural network approach uses a SmallResNet implemented in PyTorch. The network contains an initial convolutional layer followed by several residual blocks with increasing numbers of feature channels. The network ends with batch normalization, a rectified linear unit, global average pooling, dropout, and a linear layer that produces predictions for the 10 classes.

For each training data size and random seed, the three classical machine learning classifiers are trained separately, while one SmallResNet model is trained for the same condition. The experiment uses three random seeds to repeat the training process. In the statistical analysis, the results of the three classical machine learning classifiers are grouped under the Machine Learning approach, while the SmallResNet results are grouped under the Deep Learning approach.

3.5 Experimental Design

The experiment is designed to study the effects of training data size and image noise on the performance of the two approaches. The same test set is used throughout the experiment, while the training set is changed according to different data sizes. For each data size, the training samples are selected separately for each of the 10 classes to keep the class distribution balanced. Three random seeds are used to repeat the experiments and reduce the effect of random sampling.

Training Data Size Experiment. Five training data sizes are considered: 5%, 10%, 25%, 50%, and 100% of the available training set. For each size and random seed, the selected data are used to train the three classical machine learning models and the SmallResNet model. The performance of the models is then evaluated on the same test set without noise. This experiment is used to exam-

ine how the relative performance of the two approaches changes as more training data become available and to test Hypothesis 1.

Noise Experiment. To study robustness to image noise, four noise levels are applied to the test images: 0, 12.75, 38.25, and 76.5. The noise is generated from a normal distribution and added to the pixel values before the values are clipped to the valid image range. The same noisy test images are then used to evaluate both approaches. For each training data size and random seed, the trained models are evaluated at all four noise levels. This experiment is used to examine how model accuracy changes as noise increases and to test Hypothesis 2.

4 Experimental Settings

4.1 Dataset

The experiments use the CIFAR-10 dataset, which contains images from 10 classes. To reduce the computational cost while keeping the class distribution balanced, a subset of the original dataset is created. For each class, 1,000 images are randomly selected from the original training set and 200 images are randomly selected from the original test set. The resulting dataset contains 10,000 training images and 2,000 test images. The same test set is used throughout all experiments.

4.2 Data Preparation

The two approaches use different representations of the same images. For the classical machine learning approach, image features are extracted using Histogram of Oriented Gradients. The method uses 9 orientation bins, a cell size of 4×4 pixels, 2×2 cells in each block, and L2-Hys block normalization. The extracted features are then standardized before Principal Component Analysis is applied to reduce their dimensionality to 50 components.

For the convolutional neural network approach, the original color images are used directly as model input. The images are converted to tensors, rearranged into the channel-first format, and normalized by dividing pixel values by 255 so that they are represented in the range from 0 to 1.

4.3 Training Configuration

The classical machine learning approach uses three classifiers: Linear Support Vector Machine, Logistic Regression, and Random Forest. The Linear Support Vector Machine is trained with a maximum of 2,000 iterations, while Logistic Regression uses a maximum of 1,000 iterations. The Random Forest model contains 100 trees. All three classifiers use the corresponding random seed of each experiment. Principal Component Analysis is fixed at 50 components for the classical machine learning pipeline.

The convolutional neural network approach uses the SmallResNet architecture implemented in PyTorch. The network contains an initial convolutional layer followed by six residual blocks, with the number of feature channels increasing from 8 to 64. The final layers use batch normalization, a rectified linear unit, global average pooling, dropout with a rate of 0.2, and a linear classification layer for the 10 classes.

The SmallResNet is trained for 10 epochs with a batch size of 64. The Adam optimizer is used with a learning rate of 0.001, and cross-entropy loss is used as the training objective.

4.4 Training Data Sizes

Five training data sizes are evaluated: 5%, 10%, 25%, 50%, and 100% of the available training set. For each data size, the same proportion is sampled independently from each of the 10 classes to maintain a balanced class distribution. Each condition is repeated with three random seeds: 42, 43, and 44.

This design allows the study to examine model performance from a strongly limited-data setting to the full training-data setting while reducing the effect of a single random sample.

4.5 Noise Levels

Four noise levels are applied to the test images: 0, 12.75, 38.25, and 76.5. At each level, random noise generated from a normal distribution is added independently to the pixel values. The resulting values are then clipped to the valid pixel range from 0 to 255. The same noisy images are used to evaluate both approaches at each noise level. The four levels represent increasing noise strength, from clean images to strongly corrupted images. For the noise

experiment, the noise is added only to the test images, while the training data remain unchanged.

4.6 Evaluation Metrics

Accuracy is used as the main evaluation metric for both approaches. It is calculated as the proportion of correctly classified test images among all test images. The same accuracy measure is used for all training data sizes, noise levels, models, and random seeds, allowing their results to be compared under the same metric.

4.7 Statistical Analysis

A paired t-test is used to determine whether the performance difference between the two approaches is statistically significant. The results are paired by random seed and training data size so that each comparison is made under the same experimental condition. A significance level of 0.05 is used.

For Hypothesis 1, the analysis uses the results obtained with clean test images and compares the two approaches separately at each training data size. The mean accuracy of the three classical machine learning models is first grouped by training data size, noise level, and random seed. These results are then paired with the corresponding SmallResNet results before the paired t-test is applied.

For Hypothesis 2, the analysis focuses on the full training-data condition. The overall degradation of each approach is measured from noise level 0 to 76.5 by calculating the change in accuracy per unit of noise. The resulting degradation values for the three random seeds are then compared between the two approaches using a paired t-test. In addition, accuracy loss is analyzed separately over three noise intervals to examine whether the overall trend is consistent across different noise ranges.

5 Experimental Results

5.1 Performance across Training Data Sizes

The learning curve shows how the performance of the two approaches changes as the amount of training data increases.

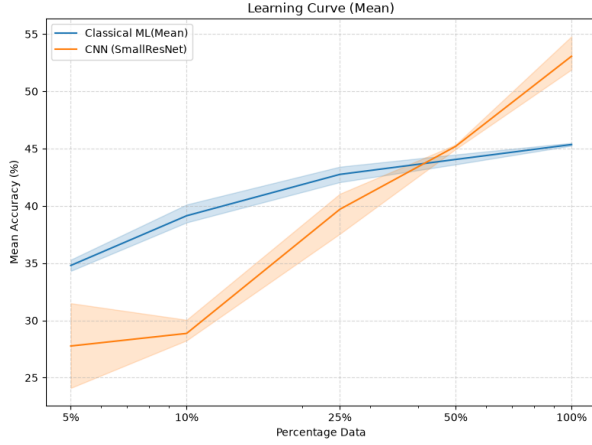


Figure 1: Mean classification accuracy of Classical ML and CNN across different training data sizes.

The classical machine learning approach performs better with 5% and 10% of the training data, with differences of 7.03 and 10.27 percentage points, respectively. As more data are used, the gap becomes smaller, and the convolutional neural network performs better at 50% and 100% of the training data. A statistically significant advantage is observed only at 100%, where the convolutional neural network outperforms the classical machine learning approach by 7.71 percentage points. These results do not support Hypothesis 1.

Table 1: Paired *t*-Test Results and Commitment Verification across Data Sizes.

Training data	Classical ML	CNN	Difference	t-statistic	p-value
5%	34.80%	27.77%	+7.03%	3.06	0.0920
10%	39.14%	28.87%	+10.27%	10.65	0.0087
25%	42.75%	39.70%	+3.05%	3.76	0.0640
50%	44.06%	45.20%	-1.14%	-4.25	0.0512
100%	45.36%	53.07%	-7.71%	-7.35	0.0180

5.2 Performance under Different Noise Levels

The robustness curve shows how the accuracy of the two approaches changes as the noise level increases when the full training set is used.

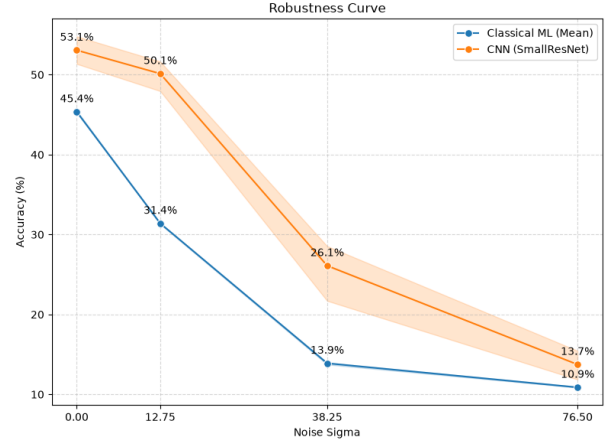


Figure 2: Classification accuracy of Classical ML and CNN across different image noise levels.

The overall analysis shows a larger degradation slope for the convolutional neural network than for the classical machine learning approach, and the difference is statistically significant (Table 2). However, the interval-based analysis (Table 3) shows that the classical machine learning approach loses much more accuracy under mild noise, while the convolutional neural network is more resistant to this initial noise level. Therefore, the overall slope does not fully represent the behavior of the two approaches across all noise levels. These results do not support Hypothesis 2.

Table 2: Overall Degradation Slope Paired *t*-Test (at 100% Data)

Mean Slope ML	Mean Slope DL	Mean diff (ML - DL)	t-statistic	p-value	Significance ($p < 0.05$)
0.45	0.51	-0.06	-66.76	0.00	Yes

6 Discussion

6.1 Data Size and Model Performance

The results suggest that classical machine learning is more effective when training data are limited, while the convolutional neural network benefits more from additional data. This may be related to the use of predefined image features in the classical machine learning pipeline, while the convolutional neural network learns features

Table 3: Multi-Interval Relative Accuracy Drop Analysis

Noise Interval	ML Initial Acc (%)	ML Final Acc (%)	ML Relative Drop (%)	DL Initial Acc (%)	DL Final Acc (%)	DL Relative Drop (%)	Faster Degradation
0.0 $\rightarrow$ 12.75	45.36	31.39	30.78%	53.07	50.13	5.53%	ML drops faster
12.75 $\rightarrow$ 38.25	31.39	13.89	55.76%	50.13	26.10	47.94%	ML drops faster
38.25 $\rightarrow$ 76.5	13.89	10.87	21.72%	26.10	13.73	47.38%	DL drops faster

directly from the images.

6.2 Noise Robustness

The noise results do not support the assumption that the convolutional neural network is more sensitive to noise because it has multiple layers. In the low-noise range, it maintains its accuracy better than the classical machine learning approach. The difference between the overall slope and the interval-based results also shows that model robustness should not be judged from a single summary value.

convolutional neural network benefits from more data and achieves higher performance when the full training set is used. Under mild noise, the convolutional neural network also maintains its accuracy better. Both hypotheses are rejected by the experimental results.

7 Threats to Validity

7.1 Construct Validity

Accuracy is used as the main performance measure, and the two approaches also use different image representations. Therefore, the comparison reflects differences between the complete approaches rather than only between the classifiers.

7.2 Internal Validity

Random sampling may affect the results, especially with limited training data. Three random seeds are used to reduce this effect, although the number of repetitions is limited.

7.3 External Validity

The experiments use a subset of CIFAR-10, three classical machine learning classifiers, and one SmallResNet architecture. The results may therefore not generalize to other datasets or model architectures.

8 Conclusion

This study compared classical machine learning and a convolutional neural network under different training data sizes and noise levels. Classical machine learning performs better with limited training data, while the con-